Neutralisation Reactions: Reactions of acids and hydroxides

Background
A neutralisation reaction is when an acid reacts with a base, in this case a hydroxide. This reaction produces water and a salt.

Neutralisation reactions are very common. Every time we brush our teeth, the toothpaste, which contains a base, neutralises the acids left on our teeth by bacteria. Indigestion, which is caused by too much acid in the stomach, can be relieved by taking an antacid, which are bases in liquid or tablet form.

The general reaction equation for an acid combining with a hydroxide is:

acid + hydroxide → water + salt

When hydrochloric acid reacts with sodium hydroxide, it produces water and sodium chloride.

$$HCl + NaOH \rightarrow H_2O + NaCl$$

Aim: To investigate the neutralisation of a hydroxide with an acid.

Equipment: Test tubes x 7
Test tube rack
Universal indicator
Distilled water
Sodium hydroxide (NaOH)
Potassium hydroxide (KOH)
Hydrochloric acid (HCl)
Sulphuric acid (H_2SO_4)
Nitric acid (HNO_3)
Safety glasses
Lab coat

Method:

1. Put one cm of distilled water in a test tube and add one drop of universal indicator. What colour is the indicator in distilled water? This is the colour you are aiming for.
2. Put approximately 12 drops of sodium hydroxide in a test tube.
3. Add <u>one</u> drop of universal indicator. Record the colour in the results table.
4. One drop at a time, add the hydrochloric acid until you notice a colour change. Record the colour in the results table.

5. Record how many drops of acid you added in the results table.
6. Repeat the process for all combinations of hydroxide and acid.

Results:

Results for Sodium hydroxide

	No. of Drops of NaOH	Initial colour of indicator	Final colour of indicator	No. of drops of acid added
HCl				
H_2SO_4				
HNO_3				

Results for Potassium hydroxide

	No. of Drops of KOH	Initial colour of indicator	Final colour of indicator	No. of drops of acid added
HCl				
H_2SO_4				
HNO_3				

Questions:

1. At what point was the solution neutral? Explain why.

2. Bees can give you a very nasty sting. The painful sting is caused by the methanoic (formic) acid they inject.
   a. What can be done to relieve the sting?

   b. Explain why vinegar would not relieve a bee sting.

3. Write a word equation for each of the reactions you conducted in your experiment.
   i. Sodium hydroxide + Hydrochloric acid →

   ii.

   iii.

   iv.

v.

vi.

Neutralisation Reactions: Reactions of acids and carbonates

Background
The reaction between an acid and a carbonate is a neutralisation reaction. These reactions produce water and a salt, as well as a third product, carbon dioxide.

The general reaction equation for an acid combining with a carbonate is:

acid + carbonate → water + salt + carbon dioxide

When hydrochloric acid reacts with sodium carbonate, it produces water, sodium chloride and carbon dioxide.

$$HCl + Na_2CO_3 \rightarrow H_2O + NaCl + CO_2$$

You can test for carbon dioxide using limewater. The limewater goes from clear to milky if carbon dioxide is present, due to the formation of a calcium carbonate precipitate.

Aim: To investigate the reaction of metal carbonates with an acid.

Equipment: Test tubes
Test tube rack
Test tube stoppers
100 mL beaker
Limewater
Pipette
10 mL measuring cylinder
Samples of various metal carbonates
Acids (hydrochloric and nitric)
Safety glasses
Lab coat

Method:

1. Pour 25 mL of limewater into the beaker. Be careful not to shake the limewater bottle.
2. Into a test tube, put a pea sized sample of sodium carbonate.
3. Measure 3 mL of hydrochloric acid in the measuring cylinder.
4. Pour the acid into the test tube and quickly stopper it.

5. Once the reaction has finished, using the pipette, add a small amount of limewater to the test tube by dribbling it down the side of the test tube.
6. Record your observations in the results table.
7. Repeat the experiment with other combinations of acids and carbonates.
8. Record all your tests and observations in the results table.

Results:

Carbonate	**Acid**	**Observation of reaction**	**Observation of limewater test**
Sodium carbonate (Na_2CO_3)	Hydrochloric (HCl)		

Questions:

Write a word equation for each of the reactions you conducted in your experiment. The first one has been started for you.

i. Sodium carbonate + Hydrochloric acid →

ii.

iii.

iv.

v.

vi.

vii.

viii.

For each of your word equations, write a formula equation underneath.

Reaction of Metals with Acids

Background

The reaction between an acid and a metal is not a neutralisation reaction because metals are not bases and the reaction does not produce water. Instead, this type of reaction produces hydrogen gas and a salt. Some metals, like sodium, react violently while some are unreactive, or inert.

The general reaction equation for an acid combining with a metal is:

acid + metal → salt + hydrogen gas

When hydrochloric acid reacts with magnesium metal, it produces magnesium chloride and hydrogen gas.

$$HCl + Mg \rightarrow MgCl_2 + H_2$$

The test for hydrogen gas is the 'pop' test. The gas produced can be collected in an inverted test tube. When a lit match is held near the mouth of the test tube, a popping sound can be heard as the hydrogen reacts with oxygen in the air to form water.

Aim: To investigate the reaction of metals with an acid.

Equipment: Test tubes
Test tube rack
Samples of various metals
Hydrochloric acid
Sulphuric acid
Acetic acid (vinegar)
Matches
Safety glasses
Lab coat

Method:

1. Pour 2 cm of hydrochloric acid into a test tube.
2. Add a small piece of magnesium to the acid.

3. If there is a noticeable reaction, hold a second test tube inverted over the reaction tube. This will collect the gases produced in the reaction.
4. Hold a lit match to the mouth of the test tube you collected the gas in. If hydrogen has been produced, you should hear a small popping sound.
5. Record your observations in the table.
6. Repeat the magnesium reaction with sulphuric and acetic acid.
7. Try other combinations of acids and metals. Record all your tests and observations.

Results:

Acid	**Metal**	**Observation of reaction**	**Hydrogen test**
Hydrochloric (HCl)	Magnesium		
Sulphuric (H_2SO_4)	Magnesium		
Acetic (Vinegar)	Magnesium		

Questions:

1. Which of the metals you tested was the most reactive?

2. Which of the metals you tested was the least reactive?

3. Write a word equation for **six** of the reactions you conducted in your experiment. The first one has been started for you.

   i. Hydrochloric acid + Magnesium →

   ii.

iii.

iv.

v.

vi.

For each of your word equations, write a formula equation underneath.